

COURSE TITLE : STATISTICAL QUALITY ASSURANCE
COURSE CODE : 5125
COURSE CATEGORY : E
PERIODS PER WEEK : 4
SEMESTER : 5
PERIODS PER SEMESTER : 52
CREDITS : 3

TIME SCHEDULE

Module	Topics	Periods
I	Quality Assurance Test	12 1
II	Statistical Quality Assurance Test	12 1
III	Control Charts Test	12 1
IV	Acceptance sampling and Control charts Test	12 1
Total		52

Course outcome :

Module	G.O	Student will be able to:
1	1	Understand Quality, Quality Control And Assurance
2	1	Understand Statistical Quality Control
3	1	Understand Control Chart.
4	1	Understand Acceptance Sampling And O C Curve

Specific outcome

MODULE I

1.1.0 Understand Quality, Quality Control And Assurance

- 1.1.1 Define Quality
- 1.1.2 State Quality Control
- 1.1.3 Explain objectives in quality control.
- 1.1.4 Explain Stages in Quality control
- 1.1.5 Explain Total Quality control
- 1.1.6 Define quality assurance.
- 1.1.7 Explain its advantageous
- 1.1.8 Explain Stages in Quality Assurance.
- 1.1.9 Explain Quality Plan.
- 1.1.10 Explain Quality Assurance Programme.

- 1.1.11 State objectives of Quality Assurance Programme.
- 1.1.12 Explain Quality Materials management.
- 1.1.13 Explain Vendor/supplier roll.

MODULE II

2.1.0 Understand Statistical Quality Control

- 2.1.1 Define statistics.
- 2.1.2 Define Statistical Quality Control.
- 2.1.3 Explain Characteristics of SQC.
- 2.1.4 Define Attributes and Variables.
- 2.1.5 Explain Process of SQC.
- 2.1.6 Construct Histogram, Frequency Polygon, Frequency Bar Chart.
- 2.1.7 Define Central Tendency.
- 2.1.8 Mean. Median. Mode.
- 2.1.9 Define Range, Standard Deviation. Variance
- 2.1.10 Compute Mean. Median. Mode. Range, Standard Deviation. Variance
- 2.1.11 Define Probability.
- 2.1.12 Explain Normal Distribution.
- 2.1.13 Construct Normal Curve.
- 2.1.14 Explain Normal Test
- 2.1.15 Explain Sampling Inspection.

MODULE III

3.1.0 Understand Control Chart.

- 3.1.1 Define Control Chart.
- 3.1.2 Explain $\bar{X}$ and R Chart.
- 3.1.3 Construct $\bar{X}$ and R Chart.
- 3.1.4 Explain Moving range Chart.
- 3.1.5 Explain P Chart.
- 3.1.6 Construct P Chart.
- 3.1.7 Construct P_n/nP , U chart.

MODULE- IV

4.1.0 Understand Acceptance Sampling and O C Curve

- 4.1.1 Define Acceptance sampling.
- 4.1.2 Explain Principle of Lot Sampling.
- 4.1.3 Explain Limitation of sampling.
- 4.1.4 Compare acceptance sampling and control chart.
- 4.1.5 Explain control charts.
- 4.1.6 Explain its benefits.
- 4.1.7 Define single sampling Plan.
- 4.1.8 Define Double sampling plan.

- 4.1.9 Define O C Curve.
- 4.1.10 Explain LTPD and AOQ.
- 4.1.11 Explain Reliability

CONTENT DETAILS

MODULE- I:

QUALITY ASSURANCE. Quality-Quality Control-Total Quality Control-Quality Assurance-Forms of Quality Assurance-stages-Quality Planning-Quality Assurance Programme-Objectives-Quality in materials management. .

MODULE- II

Statistical Quality Control.- Definition of SQC-Characteristics-attributes and variables- process of SQC-Frequency distribution-Frequency Polygon-Frequency Curve-Measure of central Tendency-Mean-Median-mode-Probability-Normal curve-Regression analysis- Normal Test-x2 Test-Comparison between 100% inspection and sampling Inspection.

MODULE- III

Control Charts. Control charts for Variables-X and R charts-calculation of 3σ Limits- Construction of X and r Charts-Moving Range charts- Accuracy and Precision-P chart-Pn/nP chart/U chart..

MODULE- IV:

ACCEPTANCE SAMPLING AND OC CURVE. Acceptance Sampling-Principle of Lot sampling-Limitations of sampling-Comparison between Acceptance sampling and control Charts-Benefits-Single sampling plan-double sampling Plan-OC Curve-Lot Tolerance Percent Defective(LTPD)-AOQ ,Reliability-Simple Problems

REFERENCE

- 1 M.Mahajan. Statistical Quality control ,S Chand
- 2 K.C.Aroa,.Total Quality Management ,Dhanapat